Online Constrained Control of Storage Systems via Simplex Disturbance-Action Policies

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***Abstract*—We study online control of a scalar storage system under nonnegative adversarial arrivals, convex state–action costs, and state-dependent action constraints. At each time, the controller chooses a feasible action before the current arrival and cost are revealed. We introduce *Simplex Disturbance-Action Control* (SDAC), a simplex-constrained finite-memory policy class whose fixed policies satisfy the per-step action constraint by construction. For online adaptation, an entropic online mirror descent (OMD) update maintains the SDAC parameters on the subprobability simplex, while scalar action clipping ensures feasibility of the actual time-varying trajectory. We prove $O\left(\sqrt{T\log(H+1)}\right)$ regret against the best fixed SDAC policy, where H is the memory horizon. We further show that SDAC geometrically approximates every feasible proportional state-feedback policy; choosing $H=\Theta(\log T)$ yields $O\left(\sqrt{T\log\log T}\right)$ regret against the best policy in this class. The setting is motivated by energy-harvesting battery management and related storage systems.**

***Index Terms*—Online control, disturbance-action policies, storage systems, energy harvesting, regret minimization, state-dependent action constraints.**

I. INTRODUCTION

A. Motivation and Problem Setting

MANY dynamic resource systems can be modeled by a storage state that evolves over time and constrains the control action at every time. The controller must choose an action subject to a state-dependent action constraint before observing the corresponding arrival and cost.

This creates a constrained discrete-time control problem with state–action coupling: aggressive early actions can harm later performance, while overly conservative actions can increase the stage cost. Our objective is to remain feasible at every time while competing with the best fixed policy in a tractable class that approximates feasible proportional state-feedback policies.

B. Model and Information Structure

We study a discrete-time storage system with state $x_t \geq 0$, where x_t models the storage level at the beginning of time t. At each time t, the controller chooses a control action u_t under the state-dependent action constraint

$$0 \leq u_t \leq \alpha x_t,$$

where $\alpha \in (0,1)$ is a fixed known retention coefficient (so a fraction $1-\alpha$ of the state is lost each period). After the action is chosen, a nonnegative adversarial arrival w_t (a resource inflow) is revealed, and the state evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t.$$

Although this input is commonly called a disturbance in the online-control literature, we call $w_t \in [0,1]$ an *arrival* to emphasize its nonnegativity; we retain the standard names Disturbance-Action Control (DAC) and Simplex Disturbance-Action Control (SDAC). Nonnegativity is essential to the storage interpretation and to the feasibility properties of SDAC.

Unlike stochastic models, our formulation is adversarial and non-anticipatory. At time t, the environment may select w_t and c_t using the history through time $t-1$, but not the current action; the controller then chooses u_t, after which w_t and the entire function c_t are revealed. The goal is low cumulative cost relative to the best fixed policy in a class defined below, with both the online and benchmark trajectories remaining feasible.

C. Application Domains

This abstraction captures a broad set of domains:

- **Energy harvesting and battery management:** x_t is battery state of charge (retention coefficient α after self-discharge), w_t is harvested energy, and u_t is discharge or consumption bounded by available energy αx_t.
- **Queueing networks and buffer management:** x_t is queue/buffer occupancy, w_t is packet arrivals, and u_t is service bounded by αx_t.
- **Inventory control:** x_t is on-hand inventory (retention coefficient α after spoilage/deterioration), w_t is supply or replenishment, and u_t is sales bounded by αx_t.
- **Water-reservoir management:** x_t is stored water (retention coefficient α after evaporation/seepage), w_t is inflow, and u_t is controlled release bounded by αx_t.

We take energy harvesting as the running interpretation; the other domains share the same storage dynamics and state-dependent action constraint.

D. Contributions

Our contributions are as follows:

- **Policy class.** We introduce SDAC, a DAC-inspired policy class for storage dynamics with the state-dependent action constraint $0 \leq u_t \leq \alpha x_t$. We prove that every fixed SDAC policy is feasible, its induced loss is convex and Lipschitz in the policy parameter, and the class geometrically approximates feasible proportional state-feedback policies.

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- **Online control and regret.** We develop an entropic OMD controller over SDAC parameters. A Bregman projection maintains the simplex constraint on the parameters, and scalar action clipping maintains feasibility of the actual trajectory. The controller achieves $O\left(\sqrt{T\log(H+1)}\right)$ regret against the best fixed SDAC policy and, with $H=\Theta(\log T)$, achieves $O\left(\sqrt{T\log\log T}\right)$ regret against the best feasible proportional state-feedback policy.

E. Related Work

a) Online nonstochastic control: A recent line of work studies online control of linear dynamical systems under adversarial disturbances and costs. Its performance criterion is cumulative regret relative to a benchmark policy class chosen in hindsight, under suitable stability conditions [1]–[3]; see [4] for an overview.

A central idea in this literature is *Disturbance-Action Control* (DAC),[1] a policy parameterization that combines a stabilizing state-feedback term with a finite linear filter of past disturbances. Under standard assumptions, this representation yields tractable surrogate losses in the policy parameters and supports efficient no-regret updates. Extensions include partial observability [5] and bandit feedback [6].

Constrained online-control methods have also been developed for affine and general static state–input constraints [7], [8]. Our focus is the storage-specific constraint $0 \le u_t \le \alpha x_t$, whose feasible action set depends on the current state. Exploiting nonnegative arrivals, SDAC uses a simplex parameterization to make every fixed policy feasible; scalar action clipping preserves feasibility when the parameter changes online.

b) Energy harvesting: Lin et al. [9] study online age-of-information optimization in an energy-harvesting system with related battery dynamics. They compare with an unrestricted offline optimum through competitive ratio; their objective depends on the energy allocation but not directly on the battery state, and the current input is revealed before allocation. In our model, the controller acts before observing the current arrival and cost, the cost may depend jointly on state and action, and performance is measured by regret against a fixed policy class.

Yu and Neely [10] study expected utility maximization under i.i.d. system states. Asgari and Neely [11] express the guarantee as an $O(\sqrt{T})$ regret–battery-capacity tradeoff and improve its dependence on the decision dimension, allowing arbitrary loss sequences while retaining i.i.d. energy arrivals. Both works use action-only objectives and fixed-decision comparators. Here, arrivals are adversarial, costs depend on both state and action, and the comparator is a fixed state-responsive policy.

c) Queueing and network optimization: Queueing control provides a mathematically related perspective based on stochastic arrivals and real-time service decisions. Foundational work emphasizes throughput and queue stability [12]–[14], while stochastic network optimization also studies tradeoffs between time-average objectives and backlog [15]. Our setting differs in its adversarial arrivals, convex state–action costs, and regret benchmark against a policy chosen in hindsight.

In many queueing applications, the natural retention coefficient is $\alpha = 1$ (no abandonment or leakage), so those systems are not direct instances of the $\alpha \in (0, 1)$ model analyzed here. Section VII discusses the additional issues that arise when $\alpha \ge 1$.

Among other results, Khojastepour and Sabharwal [16] study a time-invariant robust scheduler representable as a linear filter of past arrivals. It is designed to minimize average transmit power subject to a maximum packet-delay guarantee, rather than updated online to minimize revealed general costs.

d) Inventory management: Yang et al. [17] study online procurement under inventory constraints. The controller observes the current demand and price, then procures supply while ensuring that demand is met; performance is measured by competitive ratio for aggregate procurement cost. Our controller instead chooses an outflow before observing the current inflow and cost, and permits convex state–action costs.

Hihat and Fermanian [18] consider an inventory-control formulation with potentially nonlinear and nondifferentiable state-transition functions and study base-stock policies. While that model is more general in scope, its decision structure differs from ours: as in [17], the controller chooses supply and then observes demand through the transition, whereas our controller chooses an outflow and then observes incoming supply. Our scalar linear recursion and simplex policy class yield explicit feasibility and regret guarantees; the two frameworks target different dynamics, decisions, and policy classes.

e) Water-reservoir management: Classical dam theory studies storage dynamics with random inflows and controlled releases (see, for example, [19], [20]). Relative to our setting, this literature assumes stochastic inputs and emphasizes stationary or long-run performance quantities. It provides historical context for storage control under state constraints.

F. Paper Organization

Section III formalizes the adversarial storage-control model, assumptions, and regret benchmark. Section IV introduces the SDAC policy class and summarizes its structural properties. Section V presents the online controller, Section VI develops regret guarantees, Section VII discusses possible extensions, and Section VIII concludes. Deferred proofs appear in the Appendix.

II. Notation

We write $[T] \triangleq \{1, \dots, T\}$ and $[z]_+ \triangleq \max\{z, 0\}$. States may be indexed over $[T+1]$. For $\mathbf{v} \in \mathbb{R}^n$, the i-th coordinate is v_i; in particular, the coordinates of $\mathbf{m}_t$ and $\mathbf{g}_t$ are $m_{t,i}$ and $g_{t,i}$. We write $\mathbf{1}$ for the all-ones vector (dimension is clear from context). The Euclidean inner product is denoted by $\langle \cdot, \cdot \rangle$. We write $\log$ for the natural logarithm. We set $w_\tau \equiv 0$ for $\tau \le 0$. A fixed SDAC policy with parameter $\mathbf{m}$ is written $\pi_{\mathbf{m}}$, with induced trajectories $x_t(\mathbf{m})$ and $u_t(\mathbf{m})$. A feasible proportional policy with gain k induces $x_t(k)$ and $u_t(k)$. After the online algorithm is introduced, unadorned x_t and u_t denote the realized clipped trajectory.

[1]"Disturbance-Action Control" is the standard name in the online-control literature. In this paper the underlying signal is the nonnegative arrival w_t; we retain the DAC/SDAC names for continuity with that literature.

III. Model and Problem Setup

Let T be a positive integer. Actions, arrivals, and costs are indexed by $t \in [T]$, while states are indexed by $t \in [T+1]$. The system state $x_t \in \mathbb{R}_{\geq 0}$ denotes the resource level at the beginning of time t. The initial state is given as

$$x_1 = 0. \tag{1}$$

Fix a retention coefficient $\alpha \in (0,1)$. At each time $t \in [T]$, the controller chooses a control action $u_t \in \mathbb{R}_{\geq 0}$ subject to the state-dependent action constraint

$$0 \leq u_t \leq \alpha x_t. \tag{2}$$

The system then receives an arrival $w_t \in [0,1]$ and evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t. \tag{3}$$

In particular, the terminal state x_{T+1} is well-defined. For every feasible trajectory and every $t \in [T]$,

$$0 \leq x_{t+1} \leq \alpha x_t + 1.$$

Since $x_1 = 0$, induction gives

$$0 \leq x_t \leq \frac{1-\alpha^{t-1}}{1-\alpha} \leq \frac{1}{1-\alpha}, \qquad t \in [T+1],$$

and

$$0 \leq u_t \leq \frac{\alpha}{1-\alpha}, \qquad t \in [T].$$

Define the compact state–action envelope

$$\mathcal{D}_\alpha \triangleq \left\{ (x,u) : 0 \leq x \leq \frac{1}{1-\alpha},\ 0 \leq u \leq \alpha x \right\}. \tag{4}$$

Every feasible pair (x_t, u_t) lies in $\mathcal{D}_\alpha$; the set is a uniform compact envelope and need not coincide with the exact finite-time reachable set.

After the control action is selected, a convex cost function

$$c_t : \mathbb{R}^2_{\geq 0} \to \mathbb{R}$$

is revealed, and the incurred cost is $c_t(x_t, u_t)$. We assume each c_t is convex in (x,u) and satisfies the following Lipschitz condition on $\mathcal{D}_\alpha$: there exist constants $L_x, L_u > 0$, independent of t, such that for all $(x,u), (x',u') \in \mathcal{D}_\alpha$,

$$|c_t(x,u) - c_t(x',u')| \leq L_x|x-x'| + L_u|u-u'|. \tag{5}$$

We also assume access, at every $(x,u) \in \mathcal{D}_\alpha$, to a cost subgradient

$$\boldsymbol{\zeta}_t(x,u) = \big(\zeta_t^x(x,u), \zeta_t^u(x,u)\big) \in \partial c_t(x,u)$$

satisfying

$$|\zeta_t^x(x,u)| \leq L_x, \qquad |\zeta_t^u(x,u)| \leq L_u.$$

This oracle condition holds, for example, when the Lipschitz property above extends to an open neighborhood of $\mathcal{D}_\alpha$.

The information pattern is adversarial and non-anticipatory. At time t, the environment may choose (w_t, c_t) as a function of the history through time $t-1$, but not of the current action (or current randomization). The controller chooses u_t without observing (w_t, c_t); the pair is then revealed. Subject to this timing and the stated regularity conditions, the arrival and cost sequences may be arbitrary.

a) Assumptions:

1) **Retention coefficient:** $\alpha \in (0,1)$ is fixed and known (a fraction $1-\alpha$ of the state is lost each period).
2) **Arrival bounds:** $w_t \in [0,1]$ for all $t \in [T]$. In particular, arrivals are nonnegative resource increments, in contrast to signed process-noise disturbances common in classical control.
3) **Adversarial timing:** (w_t, c_t) may depend on the history through time $t-1$ but not on the current action, and is revealed only after u_t is chosen.
4) **Cost regularity:** each c_t is convex in (x,u), obeys (5) on $\mathcal{D}_\alpha$, and admits the bounded subgradient oracle described above.
5) **Feasible policy class:** both the online policy and the benchmark policies are required to satisfy (2) under (3).

We focus on $\alpha \in (0,1)$; the case $\alpha \geq 1$ is discussed in Section VII.

Let $\{(x_t, u_t)\}_{t=1}^T$ denote the realized trajectory of the online controller. Define its cumulative cost by

$$J_T^{\mathrm{on}} \triangleq \sum_{t=1}^T c_t(x_t, u_t). \tag{6}$$

For a fixed positive integer H, let $\mathbf{m}$ range over the SDAC parameter set $\mathcal{M}_H$ (defined in Section IV). For each $\mathbf{m} \in \mathcal{M}_H$, write

$$J_T(\mathbf{m}) \triangleq \sum_{t=1}^T c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big)$$

for the cumulative cost of the induced feasible trajectory under the same arrival sequence. The regret against the best fixed SDAC policy is defined by

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \triangleq J_T^{\mathrm{on}} - \min_{\mathbf{m} \in \mathcal{M}_H} J_T(\mathbf{m}). \tag{7}$$

While the online controller may update its SDAC parameters over time, the benchmark in (7) is the best single fixed SDAC policy chosen in hindsight. The objective is to design an online policy that satisfies (2) for all $t \in [T]$ and achieves sublinear regret as defined in (7).

IV. Simplex Disturbance-Action Control (SDAC)

A. From Disturbance-Action Control to SDAC

Disturbance-Action Control (DAC) combines a stabilizing state-feedback term with a finite linear filter of past disturbances. Under standard conditions, this parameterization yields convex surrogate losses and efficient online updates. In our scalar storage model, specializing the baseline feedback term to zero gives the representative form

$$u_t = \sum_{i=1}^H \theta_i w_{t-i}, \tag{8}$$

where H is a positive integer (the memory horizon) and $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_H)^\top \in \mathbb{R}^H$ is a linear filter. We allow nonpositive indices in such sums, with the convention $w_\tau \equiv 0$ for $\tau \leq 0$.

For our constrained model, an unrestricted filter is insufficient: its action can violate (2) by exceeding αx_t or becoming negative. We therefore introduce *Simplex Disturbance-Action Control* (SDAC), which uses a simplex-constrained filter tailored to the storage dynamics. This restriction yields fixed-policy feasibility while preserving convex surrogate losses and tractable online updates.

B. SDAC Definition

Fix a memory horizon $H \in \mathbb{N}_{>0}$. Let

$$\mathcal{M}_H \triangleq \left\{\mathbf{m} \in \mathbb{R}^H_{\geq 0} : \ \|\mathbf{m}\|_1 \leq 1\right\} \tag{9}$$

be the subprobability simplex. For any $\mathbf{m} \in \mathcal{M}_H$, define the policy $\pi_{\mathbf{m}}$ by

$$u_t(\mathbf{m}) \ \triangleq \ \sum_{i=1}^{H} m_i \alpha^i w_{t-i} = \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle, \qquad w_\tau \equiv 0 \text{ for } \tau \leq 0, \tag{10}$$

where

$$\boldsymbol{\phi}_t^u \triangleq \left(\alpha w_{t-1}, \alpha^2 w_{t-2}, \ldots, \alpha^H w_{t-H}\right)^\top \in \mathbb{R}^H, \tag{11}$$

with the same convention $w_\tau \equiv 0$ for $\tau \leq 0$.

The induced state trajectory satisfies

$$x_{t+1}(\mathbf{m}) = \alpha x_t(\mathbf{m}) - u_t(\mathbf{m}) + w_t, \qquad x_1(\mathbf{m}) = 0. \tag{12}$$

The SDAC policy class is

$$\Pi_H^{\mathrm{SDAC}} \triangleq \{\pi_{\mathbf{m}} : \mathbf{m} \in \mathcal{M}_H\}. \tag{13}$$

Whenever the policy itself is not needed as an abstract object, we minimize directly over $\mathbf{m} \in \mathcal{M}_H$. The weights α^i align the arrival filter with the retention coefficient α in (3).

We now establish the two structural properties that make SDAC useful: the induced control action always satisfies the feasibility constraint (2), and the induced cost is convex and Lipschitz in $\mathbf{m}$, so that regret minimization against the best fixed SDAC policy is an instance of online convex optimization over the simplex. Both properties follow from an explicit decomposition of the state trajectory that isolates its dependence on $\mathbf{m}$.

Recall that we set $w_\tau \equiv 0$ for $\tau \leq 0$, which is used in the base-case arguments below.

Lemma 1 (State decomposition under SDAC)**.** *Fix a positive integer H and $\mathbf{m} \in \mathcal{M}_H$. For $i \in \{1, \ldots, H\}$, define*

$$r_{t,i} \triangleq \sum_{j=1}^{i} \alpha^{j-1} w_{t-j}, \qquad r_t \triangleq \sum_{j=1}^{t-1} \alpha^{j-1} w_{t-j},$$

where the second sum is empty when $t = 1$. Then, for every $t \in [T]$,

$$x_t(\mathbf{m}) = \sum_{i=1}^{H} m_i r_{t,i} + \left(1 - \mathbf{1}^\top \mathbf{m}\right) r_t. \tag{14}$$

Define

$$\boldsymbol{\phi}_t^x \triangleq \left(r_{t,1} - r_t, \ldots, r_{t,H} - r_t\right)^\top.$$

Then

$$x_t(\mathbf{m}) = r_t + \langle \boldsymbol{\phi}_t^x, \mathbf{m}\rangle, \qquad u_t(\mathbf{m}) = \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle.$$

Moreover,

$$0 \leq r_{t,i} \leq r_t \leq \frac{1}{1-\alpha},$$

and $\mathbf{m} \mapsto x_t(\mathbf{m})$ is affine.

Proof. The auxiliary quantities satisfy

$$r_{t+1} = \alpha r_t + w_t, \qquad r_1 = 0, \tag{15}$$

and, for every $i \in \{1, \ldots, H\}$,

$$\begin{aligned} \alpha r_{t,i} - \alpha^i w_{t-i} &= \sum_{j=1}^{i-1} \alpha^j w_{t-j} \\ &= r_{t+1,i} - w_t. \end{aligned} \tag{16}$$

For $i = 1$, the sum in the first line is empty.

At $t = 1$, since $w_\tau \equiv 0$ for $\tau \leq 0$,

$$\begin{aligned} r_{1,i} &= \sum_{j=1}^{i} \alpha^{j-1} w_{1-j} = 0, \\ r_1 &= 0, \\ x_1(\mathbf{m}) &= 0, \end{aligned}$$

so (14) holds. Suppose it holds at time t. Using (12) and (10),

$$\begin{aligned} x_{t+1}(\mathbf{m}) &= \alpha \left(\sum_{i=1}^{H} m_i r_{t,i} + \left(1 - \mathbf{1}^\top \mathbf{m}\right) r_t\right) \\ &\quad - \sum_{i=1}^{H} m_i \alpha^i w_{t-i} + w_t \\ &= \sum_{i=1}^{H} m_i \left(\alpha r_{t,i} - \alpha^i w_{t-i}\right) \\ &\quad + \alpha \left(1 - \mathbf{1}^\top \mathbf{m}\right) r_t + w_t. \end{aligned} \tag{17}$$

Substituting (15) and (16) into (17),

$$\begin{aligned} x_{t+1}(\mathbf{m}) &= \sum_{i=1}^{H} m_i \left(r_{t+1,i} - w_t\right) \\ &\quad + \left(1 - \mathbf{1}^\top \mathbf{m}\right)\left(r_{t+1} - w_t\right) + w_t \\ &= \sum_{i=1}^{H} m_i r_{t+1,i} + \left(1 - \mathbf{1}^\top \mathbf{m}\right) r_{t+1}, \end{aligned} \tag{18}$$

because the total coefficient of w_t in the first line is $-\mathbf{1}^\top \mathbf{m} - \left(1 - \mathbf{1}^\top \mathbf{m}\right) + 1 = 0$. This proves (14) by induction.

All summands defining $r_{t,i}$ and r_t are nonnegative. Since $w_\tau = 0$ for $\tau \leq 0$, $r_{t,i}$ is a truncation of r_t; hence

$$0 \leq r_{t,i} \leq r_t \leq \sum_{j=1}^{\infty} \alpha^{j-1} = \frac{1}{1-\alpha}.$$

Finally,

$$x_t(\mathbf{m}) = r_t + \sum_{i=1}^{H} m_i \left(r_{t,i} - r_t\right) = r_t + \langle \boldsymbol{\phi}_t^x, \mathbf{m}\rangle,$$

which is affine in $\mathbf{m}$. □

The first consequence of this decomposition is fixed-policy feasibility.

Lemma 2 (Feasibility of SDAC policies)**.** *For every* $\mathbf{m} \in \mathcal{M}_H$*, the trajectory induced by* (10) *and* (12) *satisfies*

$$0 \le u_t(\mathbf{m}) \le \alpha x_t(\mathbf{m}) \qquad \text{for all } t \in [T]. \tag{19}$$

Proof. Because m_i, α, and w_{t-i} are nonnegative,

$$u_t(\mathbf{m}) = \sum_{i=1}^{H} m_i \alpha^i w_{t-i} \ge 0.$$

For every $i \in \{1, \dots, H\}$,

$$\alpha r_{t,i} = \sum_{j=1}^{i} \alpha^j w_{t-j} \ge \alpha^i w_{t-i}. \tag{20}$$

By Lemma 1, $1 - \mathbf{1}^\top \mathbf{m} \ge 0$ and $r_t \ge 0$. Therefore,

$$\begin{aligned} \alpha x_t(\mathbf{m}) &= \sum_{i=1}^{H} m_i \alpha r_{t,i} + \alpha\big(1 - \mathbf{1}^\top \mathbf{m}\big) r_t \\ &\ge \sum_{i=1}^{H} m_i \alpha r_{t,i} \\ &\ge \sum_{i=1}^{H} m_i \alpha^i w_{t-i} = u_t(\mathbf{m}), \end{aligned} \tag{21}$$

where the second inequality uses (20). □

For fixed arrivals, both coordinates of the SDAC state–action pair are affine in $\mathbf{m}$. The following lemma records the resulting convexity and the constants used by the online update.

Lemma 3 (Convexity and Lipschitz continuity of surrogate cost)**.** *Define*

$$\ell_t(\mathbf{m}) \triangleq c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big), \qquad \mathbf{m} \in \mathcal{M}_H,$$

and

$$G \triangleq \frac{L_x}{1-\alpha} + \alpha L_u. \tag{22}$$

Then ℓ_t *is convex on* $\mathcal{M}_H$*, and for all* $\mathbf{m}, \mathbf{m}' \in \mathcal{M}_H$*,*

$$|\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}')| \le G \|\mathbf{m} - \mathbf{m}'\|_1.$$

Moreover, at every $\mathbf{m} \in \mathcal{M}_H$*, one can select* $\mathbf{g}_t \in \partial \ell_t(\mathbf{m})$ *such that* $\|\mathbf{g}_t\|_\infty \le G$*.*

Proof. Lemma 1 and (10) give

$$x_t(\mathbf{m}) = r_t + \langle \boldsymbol{\phi}_t^x, \mathbf{m} \rangle, \qquad u_t(\mathbf{m}) = \langle \boldsymbol{\phi}_t^u, \mathbf{m} \rangle. \tag{23}$$

The coefficient bounds in Lemma 1, together with $w_{t-i} \in [0, 1]$, imply

$$\|\boldsymbol{\phi}_t^x\|_\infty \le \frac{1}{1-\alpha}, \qquad \|\boldsymbol{\phi}_t^u\|_\infty \le \alpha. \tag{24}$$

Since (23) is affine in $\mathbf{m}$ and c_t is convex, ℓ_t is convex.

By (5) and (23),

$$\begin{aligned} |\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}')| &\le L_x \big|\langle \boldsymbol{\phi}_t^x, \mathbf{m} - \mathbf{m}' \rangle\big| + L_u \big|\langle \boldsymbol{\phi}_t^u, \mathbf{m} - \mathbf{m}' \rangle\big| \\ &\le \big(L_x \|\boldsymbol{\phi}_t^x\|_\infty + L_u \|\boldsymbol{\phi}_t^u\|_\infty\big) \|\mathbf{m} - \mathbf{m}'\|_1 \\ &\le \left(\frac{L_x}{1-\alpha} + \alpha L_u\right) \|\mathbf{m} - \mathbf{m}'\|_1 \\ &= G \|\mathbf{m} - \mathbf{m}'\|_1. \end{aligned} \tag{25}$$

Let (ζ_t^x, ζ_t^u) denote the components of the bounded subgradient $\boldsymbol{\zeta}_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big)$ from Section III, and set

$$\mathbf{g}_t \triangleq \zeta_t^x \, \boldsymbol{\phi}_t^x + \zeta_t^u \, \boldsymbol{\phi}_t^u. \tag{26}$$

For every $\mathbf{m}' \in \mathcal{M}_H$, the subgradient inequality and (23) yield

$$\begin{aligned} \ell_t(\mathbf{m}') &\ge \ell_t(\mathbf{m}) + \zeta_t^x \langle \boldsymbol{\phi}_t^x, \mathbf{m}' - \mathbf{m} \rangle + \zeta_t^u \langle \boldsymbol{\phi}_t^u, \mathbf{m}' - \mathbf{m} \rangle \\ &= \ell_t(\mathbf{m}) + \langle \mathbf{g}_t, \mathbf{m}' - \mathbf{m} \rangle. \end{aligned}$$

Thus $\mathbf{g}_t \in \partial \ell_t(\mathbf{m})$. Finally,

$$\begin{aligned} \|\mathbf{g}_t\|_\infty &\le |\zeta_t^x| \, \|\boldsymbol{\phi}_t^x\|_\infty + |\zeta_t^u| \, \|\boldsymbol{\phi}_t^u\|_\infty \\ &\le \frac{L_x}{1-\alpha} + \alpha L_u = G. \end{aligned} \tag{27}$$

□

Regret minimization against the best fixed SDAC policy thus becomes online convex optimization over the simplex.

C. Expressiveness: Proportional State-Feedback Approximation

A standard policy class in online control is the family of linear state-feedback policies $u_t = -K x_t$, and a foundational property of Disturbance-Action Control is that the DAC class *approximates* any strongly stable linear policy, with approximation error decaying geometrically in the memory horizon H [1], [4].

Under our sign convention, positive u_t drains storage, so proportional feedback is written $u_t = k x_t$. In our scalar model, the feasible proportional state-feedback policies are exactly those with gain $k \in [0, \alpha]$: since $x_t \ge 0$, the feasibility constraint $0 \le u_t \le \alpha x_t$ in (2) holds if and only if $k \in [0, \alpha]$. The next lemma gives an explicit approximation within the feasible class Π_H^{SDAC}; the two endpoint policies are represented exactly.

Lemma 4 (SDAC approximation of proportional state feedback)**.** *Fix* $k \in [0, \alpha]$ *and set* $\lambda_k \triangleq \alpha - k$*. Let* $\big(x_t(k), u_t(k)\big)$ *denote the trajectory of the feasible proportional policy* $u_t(k) = k \, x_t(k)$*, so that*

$$\begin{aligned} x_{t+1}(k) &= \lambda_k \, x_t(k) + w_t, \\ t &\in [T], \qquad x_1(k) = 0. \end{aligned}$$

Define $\mathbf{m}_H^\star(k) \in \mathbb{R}^H$ *by*

$$\begin{aligned} \big[\mathbf{m}_H^\star(k)\big]_1 &\triangleq \frac{k}{\alpha}, \\ \big[\mathbf{m}_H^\star(k)\big]_i &\triangleq \frac{k \, \lambda_k^{i-1}}{\alpha^i}, \qquad i = 2, \dots, H. \end{aligned}$$

This definition satisfies $\big[\mathbf{m}_H^\star(k)\big]_i \alpha^i = k \lambda_k^{i-1}$ *for every* $i \in \{1, \dots, H\}$*, where* $\lambda_k^0 \triangleq 1$*. Then* $\mathbf{m}_H^\star(k) \in \mathcal{M}_H$*, and if each* c_t *satisfies* (5)*, then for every* $\{w_t\} \subset [0, 1]$*,*

$$\begin{aligned} &\Big|\sum_{t=1}^{T} c_t\big(x_t(\mathbf{m}_H^\star(k)), u_t(\mathbf{m}_H^\star(k))\big) - \sum_{t=1}^{T} c_t\big(x_t(k), u_t(k)\big)\Big| \\ &\qquad \le \Big(L_u + \frac{L_x}{1-\alpha}\Big) \frac{k \, \lambda_k^H}{1 - \lambda_k} \, T. \end{aligned} \tag{28}$$

The proof is deferred to Appendix A.

Theorem 1 depends on the memory horizon only through $\sqrt{\log(H+1)}$. Combining that bound with Lemma 4 therefore yields a no-regret guarantee against the best feasible proportional state-feedback policy; Corollary 1 makes this statement precise.

V. Online Algorithm

We now present our online algorithm, which updates SDAC parameters by online mirror descent (OMD) with unnormalized negentropy (equivalently, the exponentiated gradient method) and a fixed step size $\eta > 0$ over the subprobability simplex $\mathcal{M}_H \subset \mathbb{R}^H$ [21]. Let

$$\Psi(\mathbf{m}) \triangleq \sum_{i=1}^{H} \big(m_i \log m_i - m_i\big), \qquad \mathbf{m} \in \mathbb{R}^H_{\geq 0}, \tag{29}$$

denote the unnormalized negentropy (with the convention $0 \log 0 \triangleq 0$). Its Bregman divergence is the generalized Kullback–Leibler divergence

$$D_\Psi(\mathbf{m}\|\mathbf{v}) \triangleq \sum_{i=1}^{H} \Big(m_i \log \tfrac{m_i}{v_i} - m_i + v_i\Big), \tag{30}$$

defined for

$$\mathbf{m} \in \mathbb{R}^H_{\geq 0}, \qquad \mathbf{v} \in \mathbb{R}^H_{>0}.$$

Terms with $m_i = 0$ are interpreted by continuity as zero in the logarithmic part.

For each time $t \in [T]$, let $\mathbf{m}_t \in \mathcal{M}_H$ be the online parameter. We initialize at

$$\mathbf{m}_1 \triangleq \frac{1}{H+1}\mathbf{1} \tag{31}$$

so that $\sup_{\mathbf{m}\in\mathcal{M}_H} D_\Psi(\mathbf{m}\|\mathbf{m}_1) \leq \log(H+1)$ (see the proof of Theorem 1). Henceforth, unadorned x_t and u_t denote the realized clipped trajectory.

a) Control-action selection: The controller first forms the nominal SDAC action

$$\widehat{u}_t \triangleq \langle \boldsymbol{\phi}^u_t, \mathbf{m}_t\rangle, \tag{32}$$

and clips it to the admissible interval $[0, \alpha x_t]$:

$$u_t \triangleq \min\{\widehat{u}_t,\, \alpha x_t\}. \tag{33}$$

The state update is

$$x_{t+1} = \alpha x_t - u_t + w_t, \qquad x_1 = 0. \tag{34}$$

By construction, $\widehat{u}_t \geq 0$ and $u_t \in [0, \alpha x_t]$; with $x_1 = 0$ and $w_t \in [0,1]$, induction on (34) yields $x_t \geq 0$ for all $t \in [T+1]$, so the online trajectory satisfies (2).

b) Surrogate cost and OMD update: After (w_t, c_t) is revealed, define the surrogate cost by

$$\ell_t(\mathbf{m}) \triangleq c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big), \qquad \mathbf{m} \in \mathcal{M}_H, \tag{35}$$

which is convex and Lipschitz by Lemma 3. Select a subgradient $\mathbf{g}_t \in \partial \ell_t(\mathbf{m}_t)$ satisfying $\|\mathbf{g}_t\|_\infty \leq G$ as in (22), whose existence and construction are given in that lemma. We define the OMD update

$$\mathbf{m}_{t+1} \triangleq \arg\min_{\mathbf{m}\in\mathcal{M}_H} \Big\{\eta\langle \mathbf{g}_t, \mathbf{m}\rangle + D_\Psi(\mathbf{m}\|\mathbf{m}_t)\Big\}. \tag{36}$$

The regret analysis in Section VI works directly with this optimization-based definition.

c) Closed-form solution: Although (36) is a constrained optimization problem, it is easy to solve analytically because Ψ is separable and the objective is linear in $\mathbf{g}_t$. As shown in Lemma 6, the minimizer in (36) is exactly the result of two elementary steps. First a multiplicative step,

$$\widetilde{m}_{t+1,i} \triangleq m_{t,i} \exp\big(-\eta g_{t,i}\big), \qquad i = 1, \ldots, H, \tag{37}$$

where $g_{t,i}$ is the i-th coordinate of $\mathbf{g}_t$. This is followed by the Bregman projection onto $\mathcal{M}_H$,

$$\mathbf{m}_{t+1} \triangleq \mathrm{Proj}^\Psi_{\mathcal{M}_H}(\widetilde{\mathbf{m}}_{t+1}) = \frac{\widetilde{\mathbf{m}}_{t+1}}{\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\}}, \tag{38}$$

where

$$\mathrm{Proj}^\Psi_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) \triangleq \arg\min_{\mathbf{m}\in\mathcal{M}_H} D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}})$$

(the scaling is nontrivial only when $\|\widetilde{\mathbf{m}}_{t+1}\|_1 > 1$). This is the form we implement in practice.

VI. Regret Analysis

The following theorem states that the online controller achieves sublinear regret against every fixed SDAC policy $\pi_{\mathbf{m}}$, where $\mathbf{m}$ is any parameter in $\mathcal{M}_H$. The regret proof has two parts: a standard online convex optimization bound for the surrogate costs ℓ_t, and a stability bound showing that the action-clipped online trajectory stays close to the SDAC policy trajectory driven by the same parameters. Some technical lemmas required for the proof are deferred to Appendix A.

Theorem 1 (Regret bound)**.** *Suppose the cost functions satisfy the regularity and bounded-subgradient assumptions of Section III. Then the OMD controller with action clipping from Section V, initialized as in* (31) *and using step size* $\eta > 0$*, satisfies*

$$\mathrm{Reg}^{\mathrm{SDAC}}_T \leq \frac{\log(H+1)}{\eta} + \eta G^2 T\Big(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\Big),$$

where $G \triangleq \frac{L_x}{1-\alpha} + \alpha L_u$ *as in* (22)*. In particular, the choice*

$$\eta^* = \sqrt{\frac{\log(H+1)}{G^2 T\big(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2}\big)}}$$

gives

$$\mathrm{Reg}^{\mathrm{SDAC}}_T \leq 2G\sqrt{\Big(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\Big) T \log(H+1)}.$$

Combining Theorem 1 with Lemma 4 yields a uniform guarantee against the best feasible proportional state-feedback policy.

Corollary 1 (No-regret against proportional state feedback)**.** *Set*

$$H_T \triangleq \max\left\{1, \left\lceil \frac{\log T}{\log(1/\alpha)} \right\rceil\right\}.$$

Run the OMD controller with action clipping using memory horizon $H = H_T$ and the step size η^* of Theorem 1. For each $k \in [0, \alpha]$, let $\big(x_t(k), u_t(k)\big)$ denote the trajectory of the proportional policy $u_t(k) = kx_t(k)$. Define

$$\mathrm{Reg}_T^{\mathrm{prop}} \triangleq J_T^{\mathrm{on}} - \min_{k \in [0,\alpha]} J_T(k),$$

where $J_T(k) \triangleq \sum_{t=1}^{T} c_t\big(x_t(k), u_t(k)\big)$. Then

$$\begin{aligned}\mathrm{Reg}_T^{\mathrm{prop}} &\le 2G\sqrt{\left(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\right)T \log(H_T + 1)} \\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right)\frac{\alpha^{H_T+1}}{1-\alpha}\, T,\end{aligned}$$

where $G = \frac{L_x}{1-\alpha} + \alpha L_u$. Since $\alpha^{H_T} \le 1/T$, the second term is $O(1)$, while $\log(H_T + 1) = O(\log\log T)$ as $T \to \infty$. Consequently,

$$\mathrm{Reg}_T^{\mathrm{prop}} = O\Big(\sqrt{T \log\log T}\Big).$$

Thus $\mathrm{Reg}_T^{\mathrm{prop}} = o(T)$ against the best proportional state-feedback policy with $k \in [0, \alpha]$.

Proof. For $\mathbf{m} \in \mathcal{M}_{H_T}$ and $k \in [0, \alpha]$, write

$$\begin{aligned}J_T(\mathbf{m}) &\triangleq \sum_{t=1}^{T} c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big), \\ J_T(k) &\triangleq \sum_{t=1}^{T} c_t\big(x_t(k), u_t(k)\big).\end{aligned}$$

Fix $k \in [0, \alpha]$, set $\lambda_k = \alpha - k$, and use the comparator $\mathbf{m}^\star_{H_T}(k)$ from Lemma 4. Then

$$\begin{aligned}J_T^{\mathrm{on}} - J_T(k) &= \left(J_T^{\mathrm{on}} - \min_{\mathbf{m} \in \mathcal{M}_{H_T}} J_T(\mathbf{m})\right) \\ &\quad + \left(\min_{\mathbf{m} \in \mathcal{M}_{H_T}} J_T(\mathbf{m}) - J_T(k)\right) \\ &\le 2G\sqrt{\left(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\right)T \log(H_T + 1)} \\ &\quad + J_T\big(\mathbf{m}^\star_{H_T}(k)\big) - J_T(k) \\ &\le 2G\sqrt{\left(\tfrac{1}{2} + \tfrac{\alpha}{(1-\alpha)^2}\right)T \log(H_T + 1)} \\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right)\frac{k\lambda_k^{H_T}}{1-\lambda_k}\, T.\end{aligned} \tag{39}$$

The first inequality uses Theorem 1; the second uses Lemma 4. Since $k \le \alpha$, $\lambda_k \le \alpha$, and $1 - \lambda_k \ge 1 - \alpha$,

$$\frac{k\lambda_k^{H_T}}{1-\lambda_k} \le \frac{\alpha^{H_T+1}}{1-\alpha}. \tag{40}$$

The right-hand side is independent of k. Taking the minimum comparator cost over $k \in [0, \alpha]$ in (39) and applying (40) proves the finite-time bound.

Finally,

$$H_T \ge \frac{\log T}{\log(1/\alpha)} \quad \Longrightarrow \quad \alpha^{H_T} \le \frac{1}{T}.$$

Also $H_T = \Theta(\log T)$, so $\log(H_T + 1) = O(\log\log T)$ as $T \to \infty$. These two estimates give the stated asymptotic rate. □

VII. Extensions

We briefly discuss three possible extensions. Establishing their full regret bounds is left for future work.

a) Retention $\alpha \ge 1$.: For $\alpha \ge 1$, choose a fixed baseline drain Kx_t such that $\bar{\alpha} \triangleq \alpha - K \in (0, 1)$, and write $u_t = Kx_t + v_t$. The residual dynamics become $x_{t+1} = \bar{\alpha}x_t - v_t + w_t$. An SDAC filter for v_t should therefore use the powers $\bar{\alpha}^i$, not α^i. If $0 \le v_t \le \bar{\alpha}x_t$, then $0 \le u_t \le (K + \bar{\alpha})x_t = \alpha x_t$. This reduction suggests an extension of the present analysis with transformed cost and Lipschitz constants; we do not pursue the details here.

b) Vector action / multi-task allocation.: Consider a single storage state x_t with a vector action $\mathbf{u}_t \in \mathbb{R}^n_{\ge 0}$ that allocates resources across n tasks, subject to $\mathbf{1}^\top \mathbf{u}_t \le \alpha x_t$, so that the total drain from storage is the sum of the task allocations. The dynamics remain $x_{t+1} = \alpha x_t - \mathbf{1}^\top \mathbf{u}_t + w_t$, and the cost $c_t(x_t, \mathbf{u}_t)$ is convex in $(x, \mathbf{u})$. The fixed-policy SDAC construction lifts by replacing the filter vector $\mathbf{m} \in \mathcal{M}_H$ with a matrix $\mathbf{M} \in \mathbb{R}^{n \times H}_{\ge 0}$ whose entries $M_{a,i}$ sum to at most one (a matrix simplex), with each row a corresponding to a task and each column i to a memory lag. OMD can then run over this matrix simplex. If the nominal total allocation exceeds αx_t, the action vector is scaled by a nonnegative factor so that $\mathbf{1}^\top \mathbf{u}_t = \alpha x_t$.

c) Multiple storages with diagonal dynamics.: Suppose the state is $\mathbf{x}_t \in \mathbb{R}^d_{\ge 0}$ and evolves as $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t - \mathbf{u}_t + \mathbf{w}_t$ with $\mathbf{A} = \mathrm{diag}(\alpha_1, \ldots, \alpha_d)$, $\alpha_i \in (0, 1)$, arrivals $\mathbf{w}_t \in [0, 1]^d$, and per-coordinate constraints $0 \le u_{t,i} \le \alpha_i x_{t,i}$. The dynamics and feasible set then decouple by coordinate, and the natural SDAC class is a product of simplices. For a jointly convex cost, OMD operates jointly on this product set; the optimization separates by coordinate only when the cost is coordinate-separable. Cross-coupled storage systems (non-diagonal $\mathbf{A}$) require additional analysis.

These constructions retain the basic simplex/OMD structure, but their complete feasibility and regret analyses require separate treatment.

VIII. Conclusion

We studied adversarial online control of a storage system under a state-dependent action constraint. Every fixed SDAC policy is feasible by construction. For the time-varying online controller, a Bregman projection maintains the SDAC parameter constraint, while scalar action clipping maintains feasibility of the actual trajectory. The resulting controller achieves $O\Big(\sqrt{T \log(H+1)}\Big)$ regret against the best fixed SDAC policy. The SDAC approximation result further gives $O\big(\sqrt{T \log\log T}\big)$ regret against the best feasible proportional state-feedback policy when $H = \Theta(\log T)$. The model applies directly to energy-harvesting battery management and related settings in which storage limits must be respected in real time. Section VII discusses possible extensions to $\alpha \geq 1$, vector allocation, and diagonal multi-storage systems.

Appendix A
Deferred Proofs

This appendix collects the deferred proof of the proportional-state-feedback approximation guarantee from Section IV, together with the OMD and stability lemmas leading up to the regret bound stated in Section VI.

Recall that we set $w_\tau \equiv 0$ for $\tau \leq 0$, which is used in several base-case arguments below.

We first prove the proportional-policy approximation guarantee stated in Section IV-C.

Proof of Lemma 4. Fix $k \in [0, \alpha]$ and set $\lambda_k \triangleq \alpha - k \in [0, \alpha]$.

Proportional-policy trajectory. Substituting $u_t(k) = kx_t(k)$ into the dynamics gives

$$x_{t+1}(k) = \lambda_k x_t(k) + w_t, \qquad x_1(k) = 0. \tag{41}$$

Unrolling (41),

$$\begin{aligned} x_t(k) &= \sum_{j=1}^{t-1} \lambda_k^{j-1} w_{t-j}, \\ u_t(k) &= k \sum_{j=1}^{t-1} \lambda_k^{j-1} w_{t-j}. \end{aligned} \tag{42}$$

Thus $x_t(k) \geq 0$, and $k \in [0, \alpha]$ implies

$$0 \leq u_t(k) = kx_t(k) \leq \alpha x_t(k).$$

Simplex membership. Each coordinate of $\mathbf{m}_H^\star(k)$ is nonnegative. If $k = 0$, then $\mathbf{m}_H^\star(0) = \mathbf{0} \in \mathcal{M}_H$. If $k = \alpha$, then $\lambda_k = 0$, so $\left[\mathbf{m}_H^\star(\alpha)\right]_1 = 1$ and $\left[\mathbf{m}_H^\star(\alpha)\right]_i = 0$ for $i = 2, \dots, H$, hence $\mathbf{m}_H^\star(\alpha) \in \mathcal{M}_H$. Now suppose $0 < k < \alpha$. Since $k = \alpha - \lambda_k$, one has $k/\alpha = 1 - \lambda_k/\alpha$, and $\lambda_k/\alpha \in (0, 1)$. Consequently,

$$\begin{aligned} \sum_{i=1}^{H} \left[\mathbf{m}_H^\star(k)\right]_i &= \frac{k}{\alpha} + \sum_{i=2}^{H} \frac{k\, \lambda_k^{i-1}}{\alpha^i} \\ &= \frac{k}{\alpha} \sum_{j=0}^{H-1} \left(\frac{\lambda_k}{\alpha}\right)^j \\ &= \left(1 - \frac{\lambda_k}{\alpha}\right) \sum_{j=0}^{H-1} \left(\frac{\lambda_k}{\alpha}\right)^j \\ &= 1 - \left(\frac{\lambda_k}{\alpha}\right)^H \leq 1. \end{aligned} \tag{43}$$

Hence $\mathbf{m}_H^\star(k) \in \mathcal{M}_H$.

Action approximation. The identity $\left[\mathbf{m}_H^\star(k)\right]_i \alpha^i = k\lambda_k^{i-1}$ (with $\lambda_k^0 \triangleq 1$) gives

$$u_t\big(\mathbf{m}_H^\star(k)\big) = k \sum_{i=1}^{H} \lambda_k^{i-1} w_{t-i}. \tag{44}$$

Define

$$e_t^u \triangleq u_t(k) - u_t\big(\mathbf{m}_H^\star(k)\big).$$

Using $w_\tau = 0$ for $\tau \leq 0$ in (42) and (44),

$$e_t^u = k \sum_{j=H+1}^{t-1} \lambda_k^{j-1} w_{t-j}, \tag{45}$$

where the sum is empty when $t \leq H + 1$. Therefore,

$$\begin{aligned} 0 \leq e_t^u &\leq k \sum_{j=H+1}^{\infty} \lambda_k^{j-1} \\ &= \frac{k\lambda_k^H}{1 - \lambda_k}. \end{aligned} \tag{46}$$

State approximation. Let

$$e_t^x \triangleq x_t\big(\mathbf{m}_H^\star(k)\big) - x_t(k).$$

Subtracting the two state recursions yields

$$e_{t+1}^x = \alpha e_t^x + e_t^u, \qquad e_1^x = 0. \tag{47}$$

Hence

$$e_t^x = \sum_{s=1}^{t-1} \alpha^{t-1-s} e_s^u. \tag{48}$$

Combining (46) and (48),

$$\begin{aligned} 0 \leq e_t^x &\leq \frac{k\lambda_k^H}{1 - \lambda_k} \sum_{s=1}^{t-1} \alpha^{t-1-s} \\ &\leq \frac{k\lambda_k^H}{(1 - \lambda_k)(1 - \alpha)}. \end{aligned} \tag{49}$$

Cost approximation. By (5), (46), and (49),

$$\begin{aligned} &\left| c_t\big(x_t(\mathbf{m}_H^\star(k)), u_t(\mathbf{m}_H^\star(k))\big) - c_t\big(x_t(k), u_t(k)\big) \right| \\ &\quad \leq L_x e_t^x + L_u e_t^u \\ &\quad \leq \frac{L_x k\lambda_k^H}{(1 - \lambda_k)(1 - \alpha)} + \frac{L_u k\lambda_k^H}{1 - \lambda_k} \\ &\quad = \left(L_u + \frac{L_x}{1 - \alpha}\right) \frac{k\lambda_k^H}{1 - \lambda_k}. \end{aligned} \tag{50}$$

Summing (50) over $t = 1, \dots, T$ and applying the triangle inequality proves (28). ☐

The remaining lemmas establish the OMD and stability estimates used in the proof of Theorem 1; they rely on the state-decomposition, feasibility, and surrogate-cost lemmas already established in Section IV (Lemmas 1, 2, and 3).

Lemma 5 (Bregman projection onto $\mathcal{M}_H$)**.** *Let D_Ψ be the Bregman divergence* (30) *of the unnormalized negentropy* (29)*. Fix $\widetilde{\mathbf{m}} \in \mathbb{R}_{>0}^H$. Then the Bregman projection of $\widetilde{\mathbf{m}}$ onto $\mathcal{M}_H$,*

$$\mathrm{Proj}_{\mathcal{M}_H}^{\Psi}(\widetilde{\mathbf{m}}) \triangleq \underset{\mathbf{m} \in \mathcal{M}_H}{\arg\min}\, D_\Psi(\mathbf{m} \| \widetilde{\mathbf{m}}),$$

is given by

$$\mathrm{Proj}_{\mathcal{M}_H}^{\Psi}(\widetilde{\mathbf{m}}) = \frac{\widetilde{\mathbf{m}}}{\max\{1, \|\widetilde{\mathbf{m}}\|_1\}}.$$

Proof. The projection objective $\mathbf{m} \mapsto D_\Psi(\mathbf{m} \| \widetilde{\mathbf{m}})$ is strictly convex, and, for $m_i > 0$,

$$\frac{\partial}{\partial m_i} D_\Psi(\mathbf{m} \| \widetilde{\mathbf{m}}) = \log \frac{m_i}{\widetilde{m}_i}. \tag{51}$$

Its unconstrained minimizer is therefore $\mathbf{m} = \widetilde{\mathbf{m}}$.

If $\|\widetilde{\mathbf{m}}\|_1 \leq 1$, this minimizer belongs to $\mathcal{M}_H$, so $\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) = \widetilde{\mathbf{m}}$. Suppose $\|\widetilde{\mathbf{m}}\|_1 > 1$. The constraint $\|\mathbf{m}\|_1 \leq 1$ is then active. Moreover, the minimizer has no zero coordinate: if $m_\ell = 0$, transferring an arbitrarily small amount of mass from a positive coordinate to coordinate ℓ has directional derivative $-\infty$, because

$$\lim_{z \downarrow 0} \log \frac{z}{\widetilde{m}_\ell} = -\infty.$$

Thus the nonnegativity constraints are inactive. With $\nu \geq 0$ denoting the multiplier of $\|\mathbf{m}\|_1 \leq 1$, stationarity and complementary slackness give

$$\log \frac{m_i}{\widetilde{m}_i} + \nu = 0, \qquad \|\mathbf{m}\|_1 = 1. \tag{52}$$

The first relation in (52) implies

$$m_i = \widetilde{m}_i e^{-\nu}.$$

Summing over i and using the second relation in (52),

$$1 = \|\widetilde{\mathbf{m}}\|_1 e^{-\nu}, \qquad e^{-\nu} = \frac{1}{\|\widetilde{\mathbf{m}}\|_1}.$$

Hence $\nu = \log \|\widetilde{\mathbf{m}}\|_1 > 0$ and

$$\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) = \frac{\widetilde{\mathbf{m}}}{\|\widetilde{\mathbf{m}}\|_1}.$$

Combining the cases $\|\widetilde{\mathbf{m}}\|_1 \leq 1$ and $\|\widetilde{\mathbf{m}}\|_1 > 1$ proves the stated formula. □

Lemma 6 (Closed-form solution of the OMD update)**.** *Fix $\mathbf{m}_t \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$, $\eta > 0$, and $\mathbf{g}_t \in \mathbb{R}^H$, and let $\widetilde{\mathbf{m}}_{t+1}$ be defined coordinatewise by* (37)*, i.e. $\widetilde{m}_{t+1,i} = m_{t,i} e^{-\eta g_{t,i}}$. Then*

$$\begin{aligned} &\operatorname*{arg\,min}_{\mathbf{m} \in \mathcal{M}_H} \Big\{ \eta \langle \mathbf{g}_t, \mathbf{m} \rangle + D_\Psi(\mathbf{m} \| \mathbf{m}_t) \Big\} \\ &\quad = \operatorname*{arg\,min}_{\mathbf{m} \in \mathcal{M}_H} D_\Psi(\mathbf{m} \| \widetilde{\mathbf{m}}_{t+1}) = \mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}_{t+1}), \end{aligned}$$

where $\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}$ is the Bregman projection of Lemma 5. Consequently, the OMD update (36) *coincides exactly with the closed-form OMD update* (37)–(38)*, and in particular $\mathbf{m}_{t+1} \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$. Thus the positive initialization* (31) *keeps every algorithm iterate strictly positive.*

Proof. Fix $\mathbf{m} \in \mathbb{R}^H_{\geq 0}$ and a coordinate i. Then

$$\begin{aligned} \eta g_{t,i}\, m_i + m_i \log \frac{m_i}{m_{t,i}} &= m_i \left(\eta g_{t,i} + \log \frac{m_i}{m_{t,i}} \right) \\ &= m_i \log \frac{m_i}{\widetilde{m}_{t+1,i}}. \end{aligned} \tag{53}$$

Substitution into the OMD objective gives

$$\begin{aligned} &\eta \langle \mathbf{g}_t, \mathbf{m} \rangle + D_\Psi(\mathbf{m} \| \mathbf{m}_t) \\ &\quad = D_\Psi(\mathbf{m} \| \widetilde{\mathbf{m}}_{t+1}) + \sum_{i=1}^{H} (m_{t,i} - \widetilde{m}_{t+1,i}). \end{aligned} \tag{54}$$

The final sum is independent of $\mathbf{m}$. Hence the two optimization problems have the same minimizer, and Lemma 5 gives (38). Since $\mathbf{m}_t > 0$ coordinatewise, both the multiplicative step and the subsequent positive rescaling preserve strict positivity. □

Lemma 7 (One-step movement of OMD)**.** *For the OMD update with step size $\eta > 0$ and subgradient $\mathbf{g}_t \in \partial \ell_t(\mathbf{m}_t)$,*

$$\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \leq \eta \, \|\mathbf{g}_t\|_\infty.$$

In particular, if $\|\mathbf{g}_t\|_\infty \leq G$, then $\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \leq \eta G$.

Proof. First-order optimality of (36), evaluated at the feasible point $\mathbf{m}_t$, gives

$$\begin{aligned} &\langle \eta \mathbf{g}_t + \nabla\Psi(\mathbf{m}_{t+1}) - \nabla\Psi(\mathbf{m}_t), \mathbf{m}_t - \mathbf{m}_{t+1} \rangle \geq 0, \\ &D_\Psi(\mathbf{m}_{t+1} \| \mathbf{m}_t) + D_\Psi(\mathbf{m}_t \| \mathbf{m}_{t+1}) \\ &\qquad \leq \eta \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1} \rangle \\ &\qquad \leq \eta \|\mathbf{g}_t\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1. \end{aligned} \tag{55}$$

The second line uses the symmetric Bregman identity.

For every $\mathbf{z} \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$ and $\mathbf{v} \in \mathbb{R}^H$,

$$\begin{aligned} \mathbf{v}^\top \nabla^2 \Psi(\mathbf{z}) \mathbf{v} &= \sum_{i=1}^{H} \frac{v_i^2}{z_i} \\ &\geq \frac{\|\mathbf{v}\|_1^2}{\sum_{i=1}^{H} z_i} \\ &\geq \|\mathbf{v}\|_1^2. \end{aligned} \tag{56}$$

The first inequality is Cauchy–Schwarz. The segment joining $\mathbf{m}_t$ and $\mathbf{m}_{t+1}$ lies in $\mathcal{M}_H \cap \mathbb{R}^H_{>0}$ by Lemma 6; integrating (56) along this segment yields

$$\begin{aligned} D_\Psi(\mathbf{m}_{t+1} \| \mathbf{m}_t) &\geq \frac{1}{2} \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2, \\ D_\Psi(\mathbf{m}_t \| \mathbf{m}_{t+1}) &\geq \frac{1}{2} \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2. \end{aligned}$$

Combining these inequalities with (55),

$$\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2 \leq \eta \|\mathbf{g}_t\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1.$$

The claim is immediate if $\mathbf{m}_{t+1} = \mathbf{m}_t$; otherwise divide by $\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1$. □

Lemma 8 (OMD regret for the surrogate losses)**.** *For every $\mathbf{m} \in \mathcal{M}_H$, the update* (36) *satisfies*

$$\sum_{t=1}^{T} \ell_t(\mathbf{m}_t) - \sum_{t=1}^{T} \ell_t(\mathbf{m}) \leq \frac{D_\Psi(\mathbf{m} \| \mathbf{m}_1)}{\eta} + \frac{\eta}{2} \sum_{t=1}^{T} \|\mathbf{g}_t\|_\infty^2. \tag{57}$$

Proof. Convexity of ℓ_t gives

$$\begin{aligned} \ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m}) &\leq \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m} \rangle \\ &= \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1} \rangle \\ &\quad + \langle \mathbf{g}_t, \mathbf{m}_{t+1} - \mathbf{m} \rangle. \end{aligned} \tag{58}$$

First-order optimality of (36), evaluated at $\mathbf{m} \in \mathcal{M}_H$, yields

$$\begin{aligned} \eta \langle \mathbf{g}_t, \mathbf{m}_{t+1} - \mathbf{m} \rangle &\leq \langle \nabla\Psi(\mathbf{m}_{t+1}) - \nabla\Psi(\mathbf{m}_t), \mathbf{m} - \mathbf{m}_{t+1} \rangle \\ &= D_\Psi(\mathbf{m} \| \mathbf{m}_t) - D_\Psi(\mathbf{m} \| \mathbf{m}_{t+1}) \\ &\quad - D_\Psi(\mathbf{m}_{t+1} \| \mathbf{m}_t). \end{aligned} \tag{59}$$

The equality is the three-point Bregman identity. Substituting (59) into (58),

$$\begin{aligned}\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m}) \le \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_t)}{\eta} - \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_{t+1})}{\eta} \\ + \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle \\ - \frac{D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t)}{\eta}.\end{aligned} \tag{60}$$

The strong-convexity calculation (56) and Hölder's inequality give

$$\begin{aligned}&\langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle - \frac{D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t)}{\eta} \\ &\le \|\mathbf{g}_t\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 - \frac{\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2}{2\eta} \\ &\le \frac{\eta}{2}\|\mathbf{g}_t\|_\infty^2.\end{aligned} \tag{61}$$

The last inequality is Young's inequality. Combining (60) and (61), and then summing over t, gives

$$\begin{aligned}\sum_{t=1}^T \ell_t(\mathbf{m}_t) - \sum_{t=1}^T \ell_t(\mathbf{m}) \le \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} \\ - \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_{T+1})}{\eta} \\ + \frac{\eta}{2}\sum_{t=1}^T \|\mathbf{g}_t\|_\infty^2.\end{aligned}$$

Since $D_\Psi(\mathbf{m}\|\mathbf{m}_{T+1}) \ge 0$, this proves (57). □

Lemma 9 (Cumulative movement)**.** *For every $t \in \{2, \dots, T\}$ and every integer i with $1 \le i \le t-1$,*

$$\|\mathbf{m}_{t-i} - \mathbf{m}_t\|_1 \le i\,\eta G.$$

Proof. By the triangle inequality,

$$\|\mathbf{m}_{t-i} - \mathbf{m}_t\|_1 \le \sum_{j=1}^i \|\mathbf{m}_{t-j} - \mathbf{m}_{t-j+1}\|_1. \tag{62}$$

Substituting $\|\mathbf{m}_{t-j} - \mathbf{m}_{t-j+1}\|_1 \le \eta G$ from Lemma 7 into (62),

$$\begin{aligned}\|\mathbf{m}_{t-i} - \mathbf{m}_t\|_1 &\le \sum_{j=1}^i \eta G \\ &= i\,\eta G.\end{aligned}$$

□

Lemma 10 (Clipping error)**.** *For $\mathbf{m} \in \mathcal{M}_H$, define*

$$\Delta_t^x(\mathbf{m}) \triangleq x_t(\mathbf{m}) - x_t. \tag{63}$$

Here $x_t(\mathbf{m}_t)$ is the counterfactual state obtained by applying the single fixed parameter $\mathbf{m}_t$ from time 1 through time t; it is not the state generated by the time-varying sequence $\mathbf{m}_1, \dots, \mathbf{m}_t$.

Define the clipping error

$$\varepsilon_t^{\mathrm{clip}} \triangleq \widehat{u}_t - u_t. \tag{64}$$

Then, for every $t \in [T]$,

$$u_t = u_t(\mathbf{m}_t) - \varepsilon_t^{\mathrm{clip}}, \qquad 0 \le \varepsilon_t^{\mathrm{clip}} \le \alpha[\Delta_t^x(\mathbf{m}_t)]_+. \tag{65}$$

Proof. The clipping rule and the identity $\min\{a,b\} = a - [a-b]_+$ give

$$\varepsilon_t^{\mathrm{clip}} = \left[\widehat{u}_t - \alpha x_t\right]_+ \ge 0. \tag{66}$$

By (10), $\widehat{u}_t = u_t(\mathbf{m}_t)$. Fixed-policy feasibility gives

$$\widehat{u}_t = u_t(\mathbf{m}_t) \le \alpha x_t(\mathbf{m}_t).$$

Using monotonicity and positive homogeneity of $[\cdot]_+$ in (66),

$$\begin{aligned}\varepsilon_t^{\mathrm{clip}} &\le \left[\alpha x_t(\mathbf{m}_t) - \alpha x_t\right]_+ \\ &= \alpha[\Delta_t^x(\mathbf{m}_t)]_+.\end{aligned} \tag{67}$$

Finally, (64) and $\widehat{u}_t = u_t(\mathbf{m}_t)$ give the first identity in (65). □

Lemma 11 (State-gap recursion)**.** *For every fixed $\mathbf{m} \in \mathcal{M}_H$ and every $t \in [T]$,*

$$\Delta_{t+1}^x(\mathbf{m}) = \alpha\Delta_t^x(\mathbf{m}) + \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t - \mathbf{m}\rangle - \varepsilon_t^{\mathrm{clip}}. \tag{68}$$

Consequently,

$$\Delta_{t+1}^x(\mathbf{m}) \le \alpha\Delta_t^x(\mathbf{m}) + \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t - \mathbf{m}\rangle, \tag{69}$$

$$\Delta_{t+1}^x(\mathbf{m}) \ge \alpha\Delta_t^x(\mathbf{m}) + \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t - \mathbf{m}\rangle - \alpha[\Delta_t^x(\mathbf{m}_t)]_+. \tag{70}$$

Proof. The two trajectories use the same arrival w_t. Hence

$$\begin{aligned}\Delta_{t+1}^x(\mathbf{m}) &= \alpha\Delta_t^x(\mathbf{m}) - u_t(\mathbf{m}) + u_t \\ &= \alpha\Delta_t^x(\mathbf{m}) - u_t(\mathbf{m}) + u_t(\mathbf{m}_t) - \varepsilon_t^{\mathrm{clip}} \\ &= \alpha\Delta_t^x(\mathbf{m}) + \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t - \mathbf{m}\rangle - \varepsilon_t^{\mathrm{clip}}.\end{aligned} \tag{71}$$

This proves (68). The two bounds follow from $0 \le \varepsilon_t^{\mathrm{clip}} \le \alpha[\Delta_t^x(\mathbf{m}_t)]_+$ in Lemma 10. □

Lemma 12 (State deviation bound)**.** *For every $t \in [T]$, with $\Delta_t^x(\mathbf{m})$ as in (63),*

$$-\frac{\alpha\,\eta G}{(1-\alpha)^3} \le \Delta_t^x(\mathbf{m}_t) \le \frac{\alpha\,\eta G}{(1-\alpha)^2}.$$

Proof. Fix $t \in [T]$. For $s < t$, define

$$\varepsilon_{s,t}^{\mathrm{drift}} \triangleq \langle \boldsymbol{\phi}_s^u, \mathbf{m}_s - \mathbf{m}_t\rangle.$$

Since $(\boldsymbol{\phi}_s^u)_i = \alpha^i w_{s-i}$,

$$\|\boldsymbol{\phi}_s^u\|_\infty \le \alpha.$$

Lemma 9 and Hölder's inequality therefore give

$$\begin{aligned}\left|\varepsilon_{s,t}^{\mathrm{drift}}\right| &\le \|\mathbf{m}_s - \mathbf{m}_t\|_1 \|\boldsymbol{\phi}_s^u\|_\infty \\ &\le \alpha\eta G(t-s).\end{aligned} \tag{72}$$

For the fixed comparator $\mathbf{m}_t$, unrolling (68) from $\Delta_1^x(\mathbf{m}_t) = 0$ gives

$$\Delta_t^x(\mathbf{m}_t) = \sum_{s=1}^{t-1} \alpha^{t-1-s}\left(\varepsilon_{s,t}^{\mathrm{drift}} - \varepsilon_s^{\mathrm{clip}}\right). \tag{73}$$

Since $\varepsilon_s^{\mathrm{clip}} \geq 0$, (72) implies

$$\begin{aligned} \Delta_t^x(\mathbf{m}_t) &\leq \sum_{s=1}^{t-1} \alpha^{t-1-s} \left|\varepsilon_{s,t}^{\mathrm{drift}}\right| \\ &\leq \alpha\eta G \sum_{s=1}^{t-1} \alpha^{t-1-s}(t-s) \\ &= \alpha\eta G \sum_{j=1}^{t-1} j\alpha^{j-1} \\ &\leq \frac{\alpha\eta G}{(1-\alpha)^2}. \end{aligned} \tag{74}$$

The last line uses $\sum_{j=1}^{\infty} j\alpha^{j-1} = (1-\alpha)^{-2}$.

Applying (65) and (74) at time s,

$$\begin{aligned} 0 \leq \varepsilon_s^{\mathrm{clip}} &\leq \alpha[\Delta_s^x(\mathbf{m}_s)]_+ \\ &\leq \frac{\alpha^2\eta G}{(1-\alpha)^2}. \end{aligned} \tag{75}$$

This use of (74) is noncircular: its derivation used only $\varepsilon_s^{\mathrm{clip}} \geq 0$. From (73), (72), and (75),

$$\begin{aligned} \Delta_t^x(\mathbf{m}_t) &\geq -\sum_{s=1}^{t-1} \alpha^{t-1-s} \left|\varepsilon_{s,t}^{\mathrm{drift}}\right| - \sum_{s=1}^{t-1} \alpha^{t-1-s} \varepsilon_s^{\mathrm{clip}} \\ &\geq -\frac{\alpha\eta G}{(1-\alpha)^2} - \frac{\alpha^2\eta G}{(1-\alpha)^2} \sum_{s=1}^{t-1} \alpha^{t-1-s} \\ &\geq -\frac{\alpha\eta G}{(1-\alpha)^2} - \frac{\alpha^2\eta G}{(1-\alpha)^3} \\ &= -\frac{\alpha\eta G}{(1-\alpha)^3}. \end{aligned} \tag{76}$$

Together, (74) and (76) prove the claim. □

Lemma 13 (Action gap induced by clipping)**.** *For all $t \in [T]$,*

$$0 \leq u_t(\mathbf{m}_t) - u_t \leq \frac{\alpha^2\,\eta G}{(1-\alpha)^2}.$$

Proof. By (65),

$$u_t(\mathbf{m}_t) - u_t = \varepsilon_t^{\mathrm{clip}} \geq 0.$$

The same bound and Lemma 12 give

$$\begin{aligned} \varepsilon_t^{\mathrm{clip}} &\leq \alpha[\Delta_t^x(\mathbf{m}_t)]_+ \\ &\leq \frac{\alpha^2\eta G}{(1-\alpha)^2}. \end{aligned}$$

□

Lemma 14 (Stability term)**.** *For all horizons T,*

$$\sum_{t=1}^{T} \Big(c_t(x_t, u_t) - \ell_t(\mathbf{m}_t)\Big) \leq \frac{\alpha\,T\eta G^2}{(1-\alpha)^2}.$$

Proof. Both state–action pairs lie in $\mathcal{D}_\alpha$. Hence (5), Lemma 12, and Lemma 13 give

$$\begin{aligned} &c_t(x_t, u_t) - \ell_t(\mathbf{m}_t) \\ &\quad\leq L_x\left|\Delta_t^x(\mathbf{m}_t)\right| + L_u\left|u_t - u_t(\mathbf{m}_t)\right| \\ &\quad\leq \frac{\alpha\eta G L_x}{(1-\alpha)^3} + \frac{\alpha^2\eta G L_u}{(1-\alpha)^2} \\ &\quad= \frac{\alpha\eta G}{(1-\alpha)^2}\left(\frac{L_x}{1-\alpha} + \alpha L_u\right) \\ &\quad= \frac{\alpha\eta G^2}{(1-\alpha)^2}. \end{aligned} \tag{77}$$

Summing (77) over $t = 1, \dots, T$ proves the claim. □

Proof of Theorem 1. Define the stability term

$$\mathrm{Stab}_T \triangleq \sum_{t=1}^{T} \left(c_t(x_t, u_t) - \ell_t(\mathbf{m}_t)\right).$$

Using $\ell_t(\mathbf{m}) = c_t(x_t(\mathbf{m}), u_t(\mathbf{m}))$, the regret decomposes as

$$\mathrm{Reg}_T^{\mathrm{SDAC}} = \mathrm{Stab}_T + \sum_{t=1}^{T} \ell_t(\mathbf{m}_t) - \min_{\mathbf{m}\in\mathcal{M}_H} \sum_{t=1}^{T} \ell_t(\mathbf{m}). \tag{78}$$

We first bound the initial Bregman divergence. Fix $\mathbf{m} \in \mathcal{M}_H$. Since $\mathbf{m}_1 = \frac{1}{H+1}\mathbf{1}$,

$$\begin{aligned} D_\Psi(\mathbf{m}\|\mathbf{m}_1) &= \sum_{i=1}^{H} m_i \log\big((H+1)m_i\big) \\ &\quad - \|\mathbf{m}\|_1 + \frac{H}{H+1} \\ &= \sum_{i=1}^{H} m_i \log m_i + \|\mathbf{m}\|_1 \log(H+1) \\ &\quad - \|\mathbf{m}\|_1 + \frac{H}{H+1}. \end{aligned} \tag{79}$$

Because $z\log z \leq 0$ for $z \in [0,1]$,

$$D_\Psi(\mathbf{m}\|\mathbf{m}_1) \leq \|\mathbf{m}\|_1 \log(H+1) - \|\mathbf{m}\|_1 + \frac{H}{H+1}. \tag{80}$$

The right-hand side is affine in $\|\mathbf{m}\|_1 \in [0,1]$, and its values at the endpoints are

$$\begin{aligned} \frac{H}{H+1} &\leq \log(H+1), \\ \log(H+1) - \frac{1}{H+1} &\leq \log(H+1). \end{aligned}$$

Here the first inequality follows from $\log z \geq 1 - z^{-1}$ for $z \geq 1$. Therefore,

$$D_\Psi(\mathbf{m}\|\mathbf{m}_1) \leq \log(H+1) \qquad \text{for every } \mathbf{m} \in \mathcal{M}_H. \tag{81}$$

Lemma 8, $\|\mathbf{g}_t\|_\infty \leq G$, and (81) imply, for every $\mathbf{m} \in \mathcal{M}_H$,

$$\begin{aligned} \sum_{t=1}^{T} \big(\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m})\big) &\leq \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} + \frac{\eta}{2}\sum_{t=1}^{T} \|\mathbf{g}_t\|_\infty^2 \\ &\leq \frac{\log(H+1)}{\eta} + \frac{\eta G^2 T}{2}. \end{aligned} \tag{82}$$

Since (82) holds uniformly over $\mathbf{m}$,

$$\sum_{t=1}^{T} \ell_t(\mathbf{m}_t) - \min_{\mathbf{m} \in \mathcal{M}_H} \sum_{t=1}^{T} \ell_t(\mathbf{m}) \leq \frac{\log(H+1)}{\eta} + \frac{\eta G^2 T}{2}. \quad (83)$$

Lemma 14 gives

$$\mathrm{Stab}_T \leq \frac{\alpha \eta G^2 T}{(1-\alpha)^2}. \quad (84)$$

Substituting (83) and (84) into (78),

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \leq \frac{\log(H+1)}{\eta} + \eta G^2 T \left(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2} \right). \quad (85)$$

The right-hand side of (85) is positive for every $\eta > 0$. Differentiating with respect to η yields

$$-\frac{\log(H+1)}{\eta^2} + G^2 T \left(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2} \right).$$

The unique minimizer is therefore

$$\eta^* = \sqrt{\frac{\log(H+1)}{G^2 T \left(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2} \right)}}. \quad (86)$$

At this value the two summands in (85) are equal, so

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \leq 2G \sqrt{\left(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2} \right) T \log(H+1)}. \quad (87)$$

□

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